

2025

The Coding Guys

macal

Macro Tracking App

AGENDA

- 01 Introduction
- 02 Project Overview
- 03 Final Project Update
- 04 Struggles and Challenges
- 05 Recorded Demo
- 06 Conclusion



TEAM LEAD

Niko Escondo
Team Lead

Managed project sprints and documentation. Designed and implemented food search/creation/calories screens, integrated food and calorie APIs, and acted as float developer to help out as needed.



BACKEND DESIGNER

Dakota Steele
Backend Designer

Designed and constructed backend architecture, database models, and authentication systems. Additionally implemented reliable data-processing logic to ensure accurate and efficient app functionality.



UI DESIGNER

Todd Parcheta
UI Designer

Designed and constructed User Interface for application home page, stats page, history page, and settings page. Created high fidelity prototypes, integrated weight apis, and used food apis on needed pages.

macal - a calorie/nutrient tracking mobile application that serves 3 primary purposes:

1. Track daily food intake, diet, and weight
2. Record historical diet and weight data
3. Showcase this data in a clean, digestible format

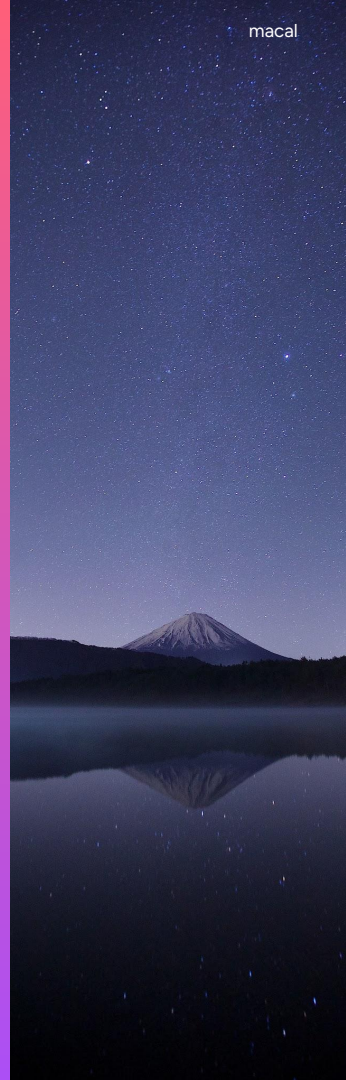
Project Overview

Final Project Update

What's Happened Since the Mid-Semester Project Update?

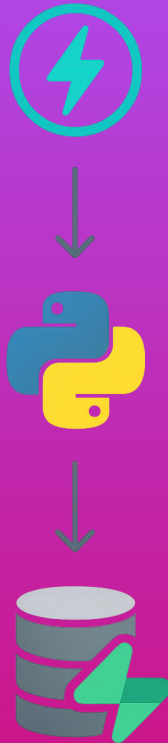
Final Project Update

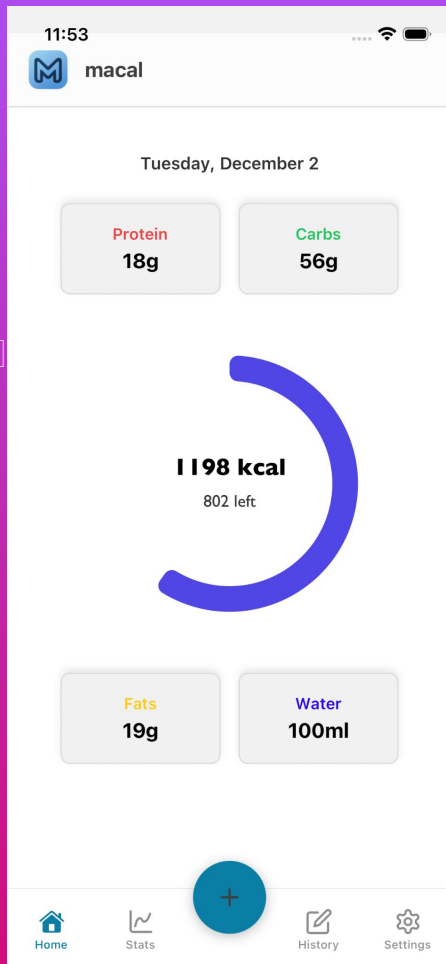
- Tailored back scope - focusing on calorie/nutrient tracking, successfully completed during project timeline
- Shifted from a web-oriented application to a mobile application
- Integrated Docker to containerize back-end and front-end environments for easier developer experience



Backend

- Handles all core logic
- Manages authentication, data processing, and API communication



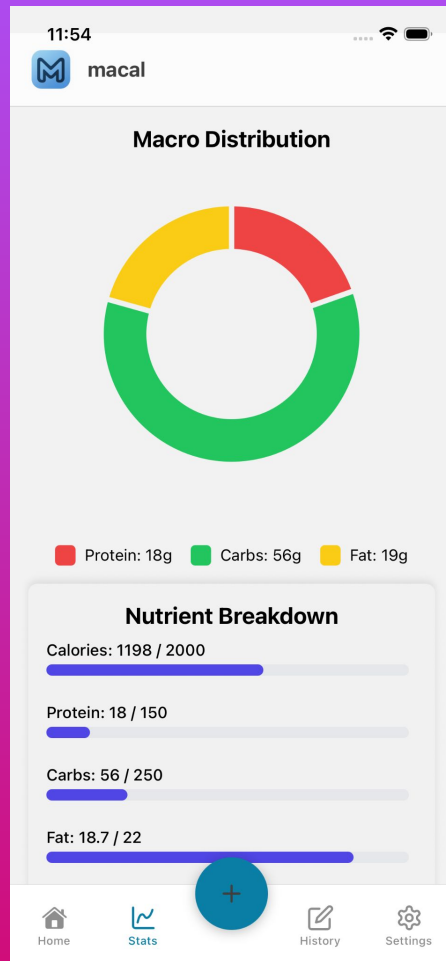


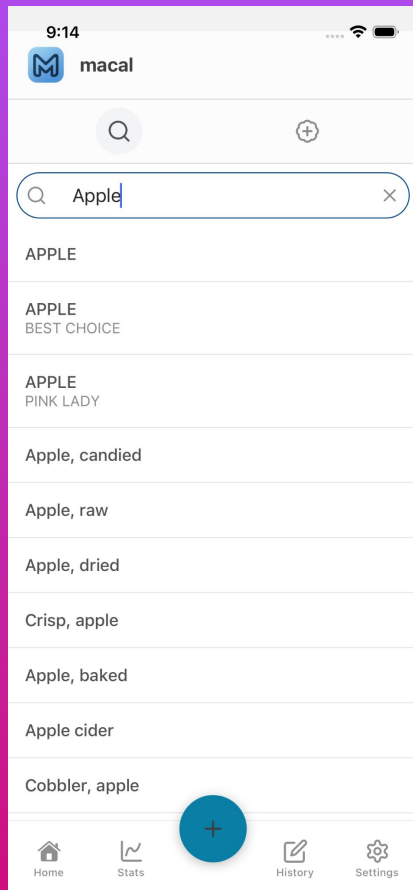
Home Page Screen

- Overview of important macros at a glance
- Updated when food is logged
- Easy to understand format

Stats Page

- In-depth overview of macros and other micro nutrients
- Tracks food intake that day
- Shows carb, protein, and fat intake in relation to one-another



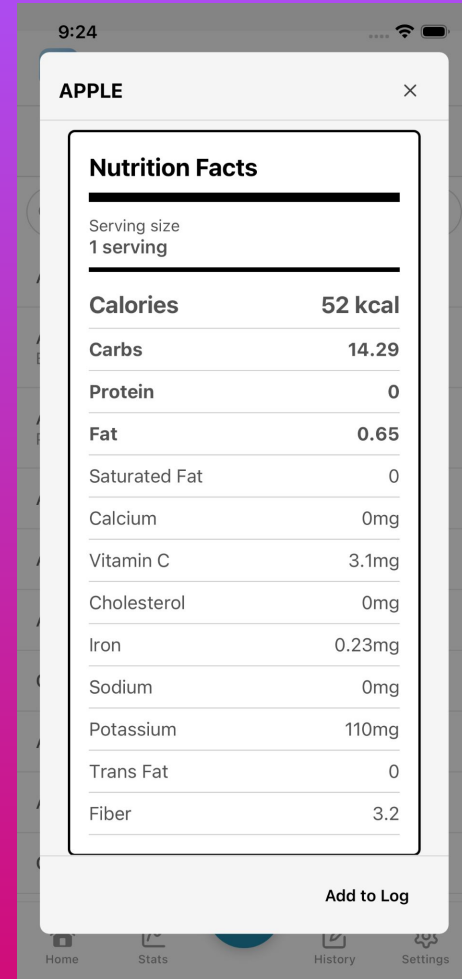


Food Search Screen

- Searching for food within the USDA Food Database
- Automatic searching every 0.5 seconds - no need to manually search

Food Search Screen - Selected Food

- Important nutrients (calories, carbohydrates, protein, and fats) sorted and prioritized first
- Nutrients followed displayed in a clean, organized manner



Food Search Screen - Create a New Food

- Logging of food's not in the USDA Food Database
- Required field and dynamic error handling

9:33 macal

Search +

Create a New Food

Food Name*

Mama's Spaghetti Recipe

Nutrition Facts

Energy (kcal)

850

Total Fat (g)

20

Trans Fat (g)

-2

ⓘ Must be a positive number

Saturated Fat (g)

Enter saturated fat (g)

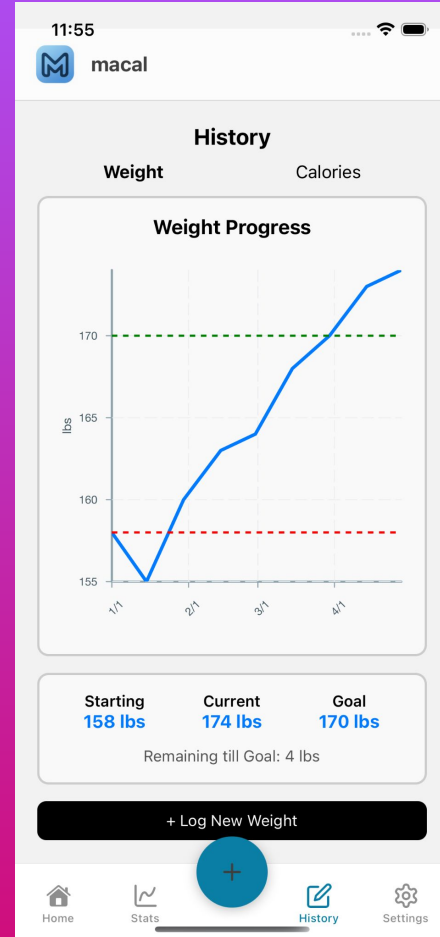
Carbohydrates (g)

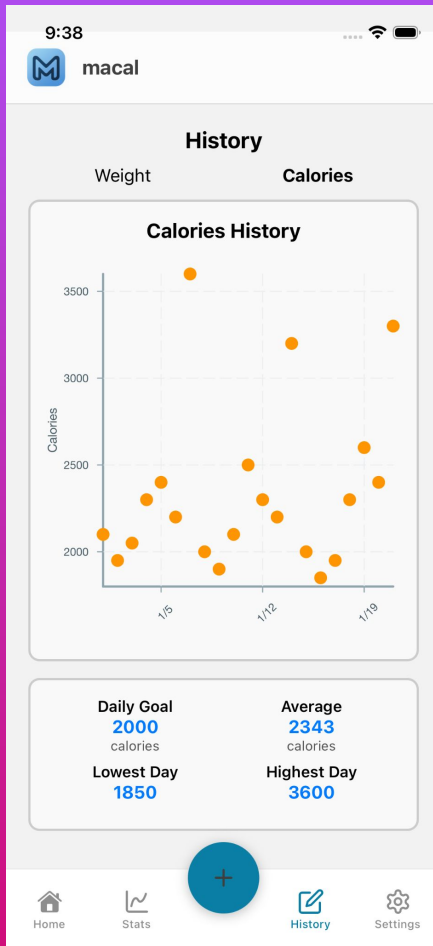
Enter carbohydrates (g)

Home Stats History Settings

History Screen - Weight Tab

- Weight tracking and visualization using VictoryChart
- Dynamically updates based on user input
- Red line = start weight
- Green line = goal weight



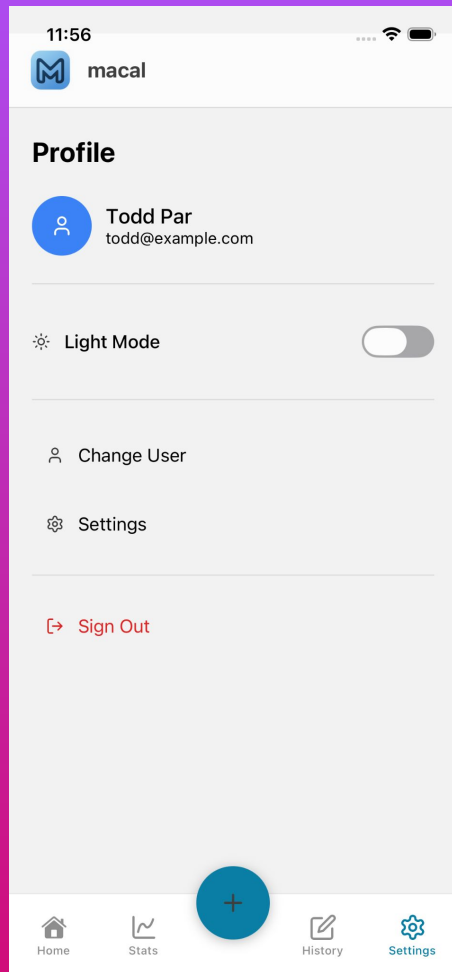


History Screen - Calories Tab

- Calorie tracking data and visualization using VictoryChart
- Dynamically updates based on user data stored in the back-end

Settings Page

- Shows current user
- Allows for signing out/
changing of accounts
- Tried having dark/light
mode



Struggles and Challenges

01

Unrealistic Expectations

An all-in-one fitness application complete with calorie/nutrient tracking, along with exercise planning and information was too much for the time (and knowledge) we had.

02

Standardizing Development Environments

A continuous struggle trying to keep up with changing project dependencies, merge conflicts, and front-end/back-end environments. Remedied with Docker.

03

Redundant Components and Functions

Creating redundant functions/components that should be reusable (ie. fetching a token) was an issue early on.

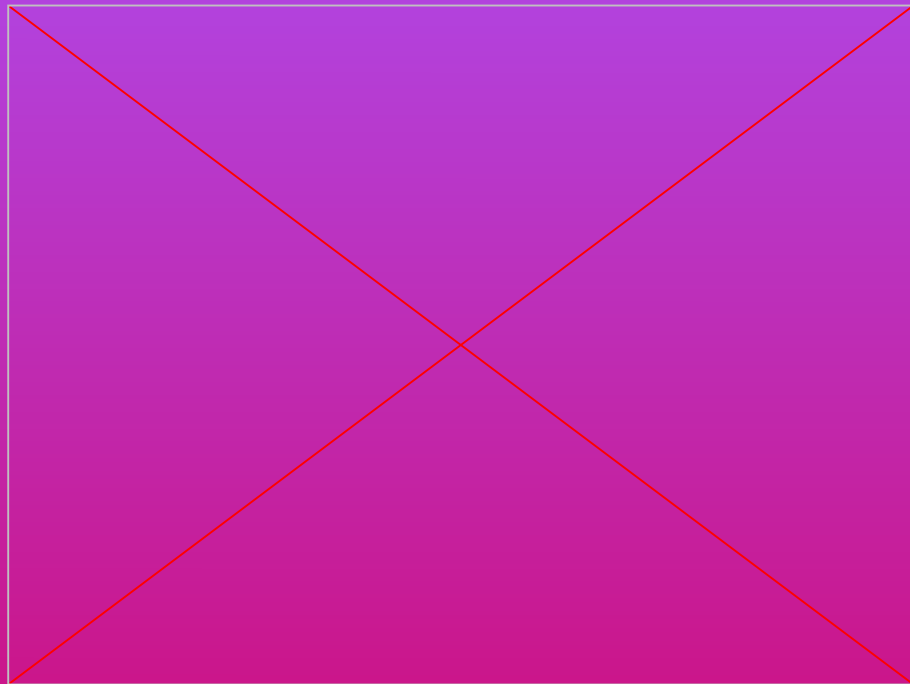
04

Time Management

Trouble with time management to reach various project and individual milestones, especially when learning new technologies, languages, and other life responsibilities was definitely a challenge.

Recorded Demo

macal in action...



THANK YOU

Any questions?